FoodBridge AI

Problem Statement

Prototype & DemoFoodBridge AI — Bridging the Gap Between Food Surplus and Food Insecurity

Large quantities of edible surplus food can remain unused at restaurants, college canteens, hotels, hostels, and events, while NGOs and community shelters need food for their beneficiaries.

Key problems

>Food waste: Surplus food may go unused because timely collection is difficult to coordinate.

>Disconnected stakeholders: Donors, NGOs, and volunteers often lack a shared coordination platform.

>Time-sensitive logistics: Food must be collected and delivered before its safety deadline.

>Capacity mismatch: Recipient organizations have different meal requirements, storage limits, and dietary policies.

>Limited transparency: Tracking donations from listing to final handover can be difficult.

Target users: Food donors, recipient NGOs and shelters, volunteer couriers, and platform administrators.

Goal: Reduce avoidable food waste by coordinating timely, suitable, and traceable food redistribution.

Proposed Approach

Execution Flow — How FoodBridge AI Works
Objective: Connect food donors with suitable NGOs through intelligent matching, coordinated delivery, and measurable impact.

1. User Registration
Donors and NGOs register and create their profiles.
Users provide relevant contact and organization details.
2. List Surplus Food
Donors enter food type, quantity, location, and expiry time.
Food safety and collection details are recorded.
3. AI-Powered Matching
The matching engine evaluates location, food type, demand, and recipient capacity.
Suitable NGOs are ranked based on compatibility and urgency.
4. Recipient Request
Eligible NGOs review available food donations.
NGOs submit requests for suitable donations.
5. Pickup & Delivery
Donors confirm recipient requests.
Collection is arranged through the donor or an assigned volunteer.
6. Completion & Feedback
Delivery status is updated after food handover.
Feedback and transaction records are maintained.
7. Analytics & Impact
The system summarizes food redistributed and meals provided.
Estimated environmental benefits and donation statistics are displayed.



Technology Stack — FoodBridge AI

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Frontend: React 19

Programming Language: TypeScript

Styling: Tailwind CSS v4

Build Tool: Vite

UI Icons: Lucide React

Charts & Analytics: Recharts

State Management: React Context API

AI Matching Engine: Rule-based, five-factor matching algorithm

Data Storage: Browser localStorage

Backend Integration: Firebase Authentication & Cloud Firestore

Testing: TypeScript-based business logic tests

Development Tools: VS Code, npm

Version Control: Git & GitHub

Why FoodBridge AI?

Our Unique Value Proposition

Smart Matching: Matches surplus food with suitable NGOs based on location, demand, food type, and capacity.

Food Safety: Tracks expiry deadlines and food safety information.

Efficient Delivery: Coordinates donors, NGOs, and volunteers for timely collection.

Transparency: Tracks donations from listing to successful delivery.

Impact Analytics: Estimates meals provided, food saved, and environmental benefits.

Our Advantage: An intelligent, capacity-aware platform that connects surplus food with people in need while reducing food waste.

Prototype & Demo







